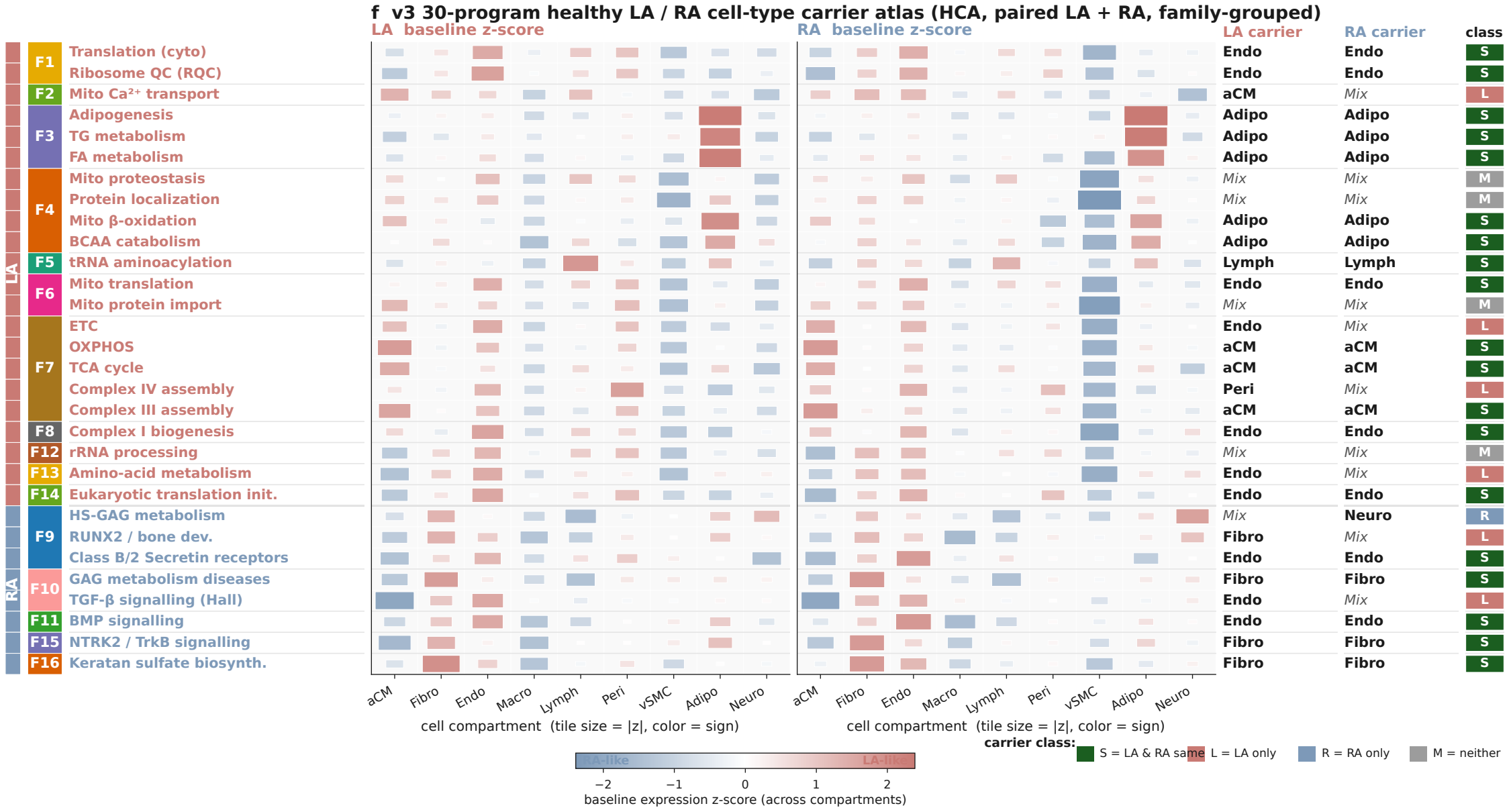

f_v3 30-program healthy LA / RA cell-type carrier atlas (HCA, paired LA + RA, family-grouped)
LA baseline z-score
RA baseline z-score
LA carrier
RA carrier
class
LA
RA
F1 Translation (cyto) Endo Endo S
Ribosome QC (RQC) Endo Endo S
F2 Mito Ca²⁺ transport aCM Mix L
F3 Adipogenesis Adipo Adipo S
TG metabolism Adipo Adipo S
FA metabolism Adipo Adipo S
F4 Mito proteostasis Mix Mix M
Protein localization Mix Mix M
Mito β-oxidation Adipo Adipo S
BCAA catabolism Adipo Adipo S
F5 tRNA aminoacylation Lymph Lymph S
F6 Mito translation Endo Endo S
Mito protein import Mix Mix M
F7 ETC Endo Mix L
OXPHOS aCM aCM S
TCA cycle aCM aCM S
Complex IV assembly Peri Mix L
Complex III assembly aCM aCM S
F8 Complex I biogenesis Endo Endo S
F12 rRNA processing Mix Mix M
F13 Amino-acid metabolism Endo Mix L
F14 Eukaryotic translation init. Endo Endo S
F9 HS-GAG metabolism Mix Neuro R
RUNX2 / bone dev. Fibro Mix L
Class B/2 Secretin receptors Endo Endo S
F10 GAG metabolism diseases Fibro Fibro S
TGF-β signalling (Hall) Endo Mix L
F11 BMP signalling Endo Endo S
F15 NTRK2 / TrkB signalling Fibro Fibro S
F16 Keratan sulfate biosynth. Fibro Fibro S
aCM Fibro Endo Macro Lymph Peri vSMC Adipo Neuro
cell compartment (tile size = |z|, color = sign)
cell compartment (tile size = |z|, color = sign)
carrier class:
S = LA & RA same
L = LA only
R = RA only
M = neither
RA-like
LA-like
−2 −1 0 1 2
baseline expression z-score (across compartments)